Report

Camera in Gazebo and ROS2

Dinh-Son Vu

Table of Contents

1. Pre-requisites

- Ubuntu 24.04 Noble can run on your computer (VM, UTM, or dual boot).
- ROS2 Jazzy is installed.
- This is simulation only. The hardware is not required.

2. Tasks

This package includes a plugin in gazebo for using a camera. In addition, it includes a fiducial marker called apriltag.

Tasks	Code
Install apriltag	sudo apt install ros-jazzy-apriltag-ros
Git clone the repo	git clone https://github.com/ROS2-rmitbot/lesson6_ws.git
Navigate to the workspace, build and source it	cd ~/lesson6_ws rm -rf build install log colcon build source install/setup.bash code .
Launch the bringup launch	ros2 launch rmit_bringup rmitbot.launch.py

Now, you can navigate in the gazebo and you may see an apriltag: this fiducial tag can be detected by the camera.

3. Explanation

Note that:

- The urdf includes a camera.xacro file, which contains a camera plugin for gz.
- The world includes and spawns aprilTag (in addition to the walls).
- The library for the AprilTag can be found at:
  https://github.com/AprilRobotics/apriltag-imgs/tree/master/tag36h11
- You may need to increase the size. In Ubuntu and in a terminal, you can use:
  # instale imagemagick
  sudo apt install imagemagick
  # convert into a larger picture
  convert tag36h11_0.png -resize 600x600 big_tag.png
- In the world, the tag is inserted with a size of 0.162m, which includes the white border. In the apriltag yaml file, the tag should fit only the black border, hence: 6/8*0.162 = 0.1215m.

4. Packages, nodes, and topics

Packages	Nodes	Description
rmitbot_bringup		Launch the other packages and nodes
rmitbot_description	display.launch.py: rviz, rsp gazebo.launch.py: gazebo	Launch the urdf and static tf
rmitbot_controller	Controller manager: jsb spawner, controller spawner	Launch input/output of the robot
rmitbot_localization	ekf	Launch the sensor fusion between wheel encoder and imu
rmitbot_mapping	slam	Launch the lidar and slam
rmitbot_navigation	nav2	Launch the navigation
rmitbot_vision	apriltag	Include the plugin for camera. Launch the apriltag detection

5. Future works

Thanks to this tag, the robot may know its relative position relative to this fixed and known position (located at [3.8, 0, 0,2]). This tag can be used later on for several purposes, namely:

- Accurate positioning of the robot for indoor environment
- Docking for charging.

The pose estimation and correction can be done with the robot_localization (ekf).